

Supervised spectral Tucker

A matched test of classification-driven factor learning

Successor to the tensor temporal screen; designed 4 October 2026.

1 Hypothesis and controlled intervention

The previous single-seed screen found that frozen spectral Tucker retained some classification performance under severe electrode loss, but its full-input validation balanced accuracy was 45.47%, below the spectral dense control’s 50.97% and the EEGNet–Transformer reference’s 63.71%. These results motivate one bounded hypothesis: updating Tucker factors for classification, while preserving observed-entry ridge inference, may improve discriminative features.

This study compares two otherwise identical spectral Tucker models. Both start with the same reconstruction-fitted factors and the same randomly initialized temporal classifier. The supervised arm allows cross-entropy gradients to update the factors; the frozen arm retains them as buffers. We use the same direct ridge-solve implementation in both arms, so a change of numerical solver is not confounded with supervision. This is factor learning during classifier training, not fine-tuning an already selected classifier checkpoint.

Property	Frozen control	Supervised factors
Spectral features and normalization	Fixed; training-calibrated	Identical
Initial Tucker factors	Training reconstruction fit	Identical fit and seed
Initial head and data order	Paired	Paired
Core inference	Observed-entry ridge solve	Same solve, differentiable
Factors during classification	Frozen buffers	Trainable U, V
Factor constraint	Initial unit columns retained	Unit columns after updates
Classification loss	Cross-entropy	Same cross-entropy
Training availability	Full 22 channels	Full 22 channels

All comparisons in this report concern these two new fits. The previous EEGNet–Transformer score is historical context, not a newly matched arm in this experiment. A gain would support task-informed factor adaptation under the tested settings, rather than proving a universal advantage of tensor models.

2 Pipeline and differentiable core

BCI IV 2a supplies four motor-imagery classes and 22 EEG electrodes at 250 Hz. We retain the verified participant-specific T-session splits and the previous artifact exclusions, cue alignment, 2–30 Hz window-local filtering and training-only channel normalization. Each trial has four 250-sample windows. Four overlapping 100-sample patches per window (stride 50) give 16 ordered time positions without crossing window or availability boundaries.

A periodic Hann window and FFT yield 12 log-power features at 2.5–30 Hz. The feature mean and population standard deviation are fitted on training trials only, separately for each channel/frequency. With patch index t , standardized features have shape $F[B, 22, 16, 12]$. Boolean availability $M[B, 22, 4]$ expands to the 16 patch positions. Hidden values are selected away before arithmetic, and every original window must retain at least one observed channel.

Reconstruction fitting initializes $U \in \mathbb{R}^{22 \times 4}$ and $V \in \mathbb{R}^{12 \times 4}$ for 30 epochs, using training data only.

For a patch, the inferred core solves

$$G^* = \arg \min_G \frac{\|F_O - U_O G V^\top\|_F^2}{12|O|} + \lambda \|G\|_F^2, \quad \lambda = 10^{-3}.$$

Let $A = U_O^\top U_O$, $B = V^\top V$, $C = U_O^\top F_O V$ and $\rho = 12|O|\lambda$. Then

$$AGB + \rho G = C.$$

Using row-major vectorization, the 16×16 system is

$$H_{(r,s),(r',s')} = A_{rr'} B_{ss'} + \rho \delta_{rr'} \delta_{ss'}, \quad g = H^{-1} c.$$

Implementation uses `torch.linalg.solve` in float64, without explicitly forming an inverse. Positive ridge makes H positive definite even when fewer than four electrodes are observed or factor columns correlate. Differentiating the solve propagates

$$dg = H^{-1}(dc - dH g)$$

to both factors. There is no detached core and no differentiation through an eigendecomposition. The frozen and supervised forward functions agree at matched initialization; the frozen result is also checked against the original solver.

Each 4×4 core is flattened to 16 values and projected to 32 features. Fixed sinusoidal positions and projected availability bits enter a one-layer temporal Transformer: four heads, feed-forward width 64, pre-LayerNorm, GELU and dropout 0.1. Final normalization, mean pooling and 88 original mask flags feed the four-class linear output. Spectral normalization remains fixed in both arms; the learned factors are the only added trainable quantities.

Unit column norms are maintained by projection after supervised optimizer updates. This is a norm constraint, not orthogonality. Reconstruction fitting remains an initialization stage; classifier training adds no reconstruction loss.

3 Training, evaluation and decision rule

The loss is unweighted four-class cross-entropy. Both arms use AdamW with learning rate 10^{-3} , weight decay 10^{-4} , batch size 32 and gradient norm clipping at 1. The same learning rate applies to the head and supervised factors. Ten warm-up epochs precede cosine decay. Fits run for at most 250 epochs; early stopping is permitted after 75 epochs with patience 50. Full-input validation balanced accuracy selects the checkpoint, breaking ties by lower log loss and then earlier epoch.

There are nine participants and three paired split/training seeds (0, 1, 2): 27 paired tasks and 54 fits. Each seed changes both the historical split and training randomness; these are not three restarts of one fixed split. No test predictions or scores enter fitting, checkpoint selection or this analysis.

Validation includes full availability and static/dynamic random retention of 16 or 6 electrodes. Each degraded condition has five paired repeats. Aggregation averages repeats, then seeds within participant, then participants equally. The primary contrast is supervised minus frozen full-input balanced accuracy. The exploratory screen requires at least two percentage points mean improvement and improvement in at least six of nine participant means. Participant bootstrap intervals use 2,000 resamples and seed 20261004. Seed-level findings, log loss, train-validation gaps and degraded conditions are secondary evidence.

Three seeds improve the assessment of split/training variation, but do not make validation independent of earlier research. An independent session or dataset would be needed for confirmation. Condition-specific intervals are exploratory and do not correct for multiple comparisons. All arms use full-input training; mixed-availability training remains a separate hypothesis.

Source/configuration/package/input hashes identify the new study. Calibration and initial-state hashes check pair matching, selected checkpoints are reloaded, and artifacts are verified before reuse. The previous studies’ source files, configurations and scientific artifacts are preserved.

4 Results and verdict

All 54 real-data fits completed (nine participants, three paired split/training seeds, two arms). The following values average seeds within participant and participants equally. Accuracy values are percentages; differences are percentage points. Log loss is unitless and lower is better.

Model	Train BA	Validation BA	Gap	Val. log loss
Frozen factors	67.44	45.71	21.73	1.593
Supervised factors	72.86	47.47	25.40	1.570

Participant	Frozen BA	Supervised BA	Difference
A01	39.74	45.70	+5.95
A02	38.10	41.80	+3.71
A03	59.71	63.42	+3.71
A04	42.36	41.93	-0.43
A05	40.38	41.03	+0.64
A06	42.42	42.58	+0.15
A07	41.85	43.04	+1.19
A08	48.53	49.08	+0.55
A09	58.32	58.62	+0.31

Availability	Frozen BA	Supervised BA	Difference
Full 22	45.71	47.47	+1.75
Static 16	37.94	40.03	+2.08
Dynamic 16	37.61	40.47	+2.86
Static 6	35.42	35.38	-0.04
Dynamic 6	35.52	36.37	+0.85

Primary finding. Updating the factors improved full-input balanced accuracy by 1.75 percentage points, with an exploratory participant-bootstrap 95% interval of [0.56, 3.18] points. Eight of nine participant averages improved; A04 declined by 0.43 points. However, the prespecified two-point mean-gain threshold was not met. The appropriate verdict is a small, fairly consistent benefit from supervised factors, with a failed practical screening criterion. It is not a null result, nor evidence of a strong classification architecture.

Split/training-seed mean gains were +1.80, +2.58 and +0.87 points for seeds 0, 1 and 2. Seventeen of the 27 individual participant/seed pairs improved, eight declined and two tied (absolute tolerance 10^{-12} in balanced accuracy). Training balanced accuracy increased by 5.42 points while validation increased by 1.75, widening the mean train-validation gap from 21.73 to 25.40 points. This is consistent with additional fitting capacity exceeding the gain in generalization; it does not uniquely identify the cause. Full-input log loss changed by -0.023 with interval $[-0.243, 0.182]$, providing no clear improvement in probabilistic prediction. Both mean full-input log losses exceed the uniform four-class predictor’s $\log 4 \simeq 1.386$, despite above-chance balanced accuracy. This indicates a weakness in confidence quality. All 54 fits stopped before the 250-epoch limit (75–121 epochs); extending the epoch budget is not supported by these histories.

Electrode-loss findings. At 16 retained channels, gains were +2.08 points for static masks (interval [0.81, 3.10]) and +2.86 for dynamic masks ([1.08, 4.47]), each positive in eight participants. At six channels, static masks changed by -0.04 points ($[-1.28, 1.45]$) and dynamic masks by $+0.85$ ($[-0.89, 2.72]$). Thus this intervention helps moderate-loss accuracy, but does not establish a severe-loss benefit. Mean log loss at six channels increased by 0.068 (static) and 0.027 (dynamic), with both intervals crossing zero. All these condition-specific findings are exploratory.

Research decision. Keep differentiable factor learning as a modestly supported component, but do not promote this model as the accuracy baseline. The 47.47% validation balanced accuracy remains below the earlier spatial Transformer’s 63.71% single-seed screen and 64.87% three-seed study; those are historical context, not matched contrasts in this study. The result supports classification-informed factor adaptation at these fixed spectral features, ranks and head, but cannot show that task adaptation alone repairs the information bottleneck. It also does not test learned raw temporal features.

A useful next bounded comparison would train both frozen and supervised arms with the same mixed-availability schedule, retaining this full-training study as a control. That would test whether exposure to missing channels improves the six-channel outcome. For the separate full-input accuracy objective, test a simpler or more strongly regularized classifier on the same cores before increasing factor ranks or Transformer depth. The larger train-validation gap motivates this test but does not prove that classifier capacity is the cause. Each change requires its own declared identity and controls; neither experiment was run here.

Verification and provenance. The independent audit replays all 54 selected checkpoints on training and validation data, reconstructs all 1,134 availability evaluations, recalibrates each task’s training-only initial factors, and independently checks aggregate means and bootstrap intervals. Selected supervised factors changed in all 27 fits, while frozen factors were unchanged; paired initial full states and calibration histories match exactly. The software checks comprise 25 new model/runner tests and 17 tensor temporal regression tests. The numerical review checks the direct system against an explicit observed-entry design and verifies gradients. Historical source/configuration/workflow/result preservation checks cover 1,122 files.

All 85 Python source hashes, 64 input hashes, configuration and package versions are recorded in the manifest. The study identity is

479db2ec18e7f1e1b5d1e5255c5a200c3df7d59b7d8a73936b4415252af8ed57

The machine-readable results and independent audit are under `results/supervised-tucker-v1/report/`. Fits used one CPU thread each; cohort execution used up to four independent processes. Elapsed times are execution records, not controlled speed benchmarks. Checkpoint selection and reporting used validation only, so an independent session or dataset remains necessary before a confirmatory accuracy claim.

5 Reproduction

Declaration: `configs/supervised-tucker.json`.

Implementation: `inm/supervised_tucker/`.

Launcher: `scripts/run_supervised_tucker_experiments.sh`.

Use separate `--pilot`, `--cohort`, `--summarize` and `--dry-run` modes. Outputs belong to `results/supervised-tucker-v1/`.

Build this report with `make -C docs/concept supervised_tucker.pdf`.